MIL-C-46952 (MU)
26 February 1963

MILITARY SPECIFICATION

CHARGE, PROPELLING, 155MM, XM115 (WHITE BAG)
LOADING, ASSEMBLING AND PACKING

1. SCOPE

1.1 This specification covers the loading, assembling and packing for one type of propelling charge designated as Charge, Propelling, 155MM, XM115 for Howitzer, T255 and T258 for use with Projectile, M470.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-223 - Powder, Black
MIL-A-2550 - Ammunition and Special Weapons; General Specification for
MIL-B-45451 - Benite Strands
MIL-I-45208 - Inspection Requirements, General Specification for
MIL-I-45607 - Inspection Equipment, Supply and Maintenance of

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-286 - Propellants: Sampling, Inspection and Testing

DRAWINGS

ORDNANCE CORPS

119154 - Increment for Loading Assembly with Base Igniter

FSC: 1320

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119249 - Charge, Propelling, 155MM, XM115 (White Bag), for Howitzer T255 and T258
8800640 - Box, Packing, Ammo, for Charge, Propelling, 155MM, M51
8800641 - Container, Ammo, Fiber, XM499, for Charge Propelling, 155MM, M51

PUBLICATIONS

ORDNANCE CORPS

ORD-M608-11 - Procedures and Tables for Continuous Sampling by Attributes
IEL-119249 - Index of Inspection Equipment List

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 Material.-Materials shall be in accordance with applicable drawings and specifications.

3.2 Propelling charge.-The propelling charge shall comply with all requirements specified on Drawing (dwg.) 119154 and 119249, and with all requirements specified in applicable specifications.

3.3 Moisture content

3.3.1 Black powder.-The moisture content of the black powder at the loading station, at the time of loading shall not exceed the amount specified in MIL-P-223, when tested as specified in 4.3.1.

3.3.2 Strand benite.-The moisture content of the strand benite at the loading station, at the time of loading shall not exceed the amount specified in Specification MIL-B-45451 when tested as specified in 4.3.1.

3.4 Propelling charge increment weight.-The charge increment weights shall comply with the weights specified in the loading authorization within the following tolerances:

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Increment number	Tolerance
	ounces
4	0.05
5	0.05
6	0.10
7	0.10

3.5 Chamber gage.-The completed propelling charge shall freely enter the prescribed chamber gage.

3.6 Workmanship.-All components shall be free of dirt, grease, and other foreign material.

4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions.-The supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 in order to define terms used herein. The provisions of Specification MIL-A-2550 shall apply.

4.1.1 Contractor quality assurance system.-The contractor shall provide and maintain an effective quality assurance system in compliance with the requirements of Specification MIL-I-45208.

4.1.2 Submission of product.-At the time the completed lot of product is submitted to the Government for acceptance the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all quality assurance provisions of the approved current written description of the system.

b. Number of units of product inspected.

c. Results obtained for all inspection performed.

d. Drawing, specification number and date, together with an identification and date of changes.

e. Certificates of analysis on all material procured directly by the contractor when such material is controlled by Government specifications

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referenced in any of the contractual documents.

f. Number of items in the lot.
g. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.1.3 Government verification.-Using the contractor's written quality assurance procedure, this detail specification, the applicable drawings and other contractual documents as a guide, the Government inspector shall verify at unscheduled intervals all quality assurance operations performed by the contractor. Verification will be in accordance with Specification MIL-I-45208 and will be performed to the extent necessary to assure compliance with the contractual requirements. Severity of Government inspection of individual characteristics will be directly related to the seriousness of the classification assigned. In no instance will a characteristic classified "critical" be accepted solely on the basis of the contractor's records.

4.2 Inspection provisions.

4.2.1 Lot formation.-A lot shall consist of propelling charges produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification, and same specification revision. Drawing, specification, and process changes not affecting safety, performance, or fit, as determined by the Government, shall not necessitate changing the lot number. Each lot shall contain:

a. Propellant of one granulation from not more than one lot.
b. Black powder of one interfix lot number from one manufacturer.
c. Benite strands of one interfix lot number from one manufacturer.

4.2.2 Examination.-Sampling plans and procedures for the following classifications of defects shall be in accordance with Standard MIL-STD-105 except that inspection for critical defects, when listed, shall be 100 percent. Continuous sampling plans in accordance with Handbook ORD-M608-11 may be used if approved by the procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans may be applied to the individual characteristics listed using an AQL of 0.25 percent for each major defect and an AQL of 0.40 percent for each minor defect. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.2.4.

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4.2.2.1 Benite cellulose nitrate igniter (see dwg. 8863420 covering a detail of 119154).

Categories	Defects	Method of Inspection	Code No. (see 6.3)
Critical:	None defined		
Major:	AQL 0.25 percent		
101.	Outside diameter	Gage	01001
102.	Total length	Gage	01002
Minor:	AQL 1.50 percent		
201.	Ties (tape) missing or improper length	Gage	01003
202.	Ties (thread) missing or improperly located	Gage	01004
203.	Lacquer missing	Visual	01005
204.	Coating missing or improper	Visual	01006

4.2.2.2 Increment No. 4 Assembly (see dwg. 119154).

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Igniter charge assembly missing	Visual	02001
2.	Benite-cellulose nitrate igniter out of position, i.e., not in contact with or not securely fastened to base igniter pad	Visual-Manual	02002
Major:	AQL 0.25 percent		
101.	Assembly damaged to the extent that black powder or propellant can escape	Visual	02003
Minor:	AQL 1.50 percent		
201.	Any seam or opening incompletely stitched	Visual	02004
202.	Thread broken	Visual	02005
203.	Evidence of poor workmanship (see 3.6)	Visual	02006
204.	Straps missing or improperly located	Visual	02007

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4.2.2.3 Increment bag loading assembly increments 5, 6 and 7 (see dwg. 119155 covering a detail of 119249).

Categories	Defects	Method of Inspection	Code No
Critical: None defined			
Major:	AQL 0.25 percent		
101.	Assembly damaged to the extent that propellant can escape	Visual	03001
Minor:	AQL 1.50 percent		
201.	Any seam or opening incompletely stitched	Visual	03002
202.	Thread broken	Visual	03003
203.	Evidence of poor workmanship (see 3.6)	Visual	03004

4.2.2.4 Propelling charge, prior to securing straps (see dwg. 119249).

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Igniter core assembly missing, or insecure	Visual-Manual	04001
Major:	AQL 0.25 percent		
101.	Assembly damaged to the extent that propellant can escape	Visual	04002
102.	Any component except igniter core assembly missing or improperly assembled	Visual	04003
Minor:	AQL 1.50 percent		
201.	Any seam or opening incompletely stitched	Visual	04004
202.	Threads broken	Visual	04005
203.	Evidence of poor workmanship (see 3.6)	Visual	04006

4.2.2.5 Propelling charge (see dwg. 119249).

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Categories	Defects	Method of Inspection	Code No.
Critical: None defined			
Major:	AQL 0.40 percent		
101.	Chamber gage failure	Gage	05001
102.	Total length, max	Gage	05002
103.	Assembly damaged to the extent that black powder, benite strands or propellant can escape	Visual	05003
104.	Color of cloth improper	Visual	05004
Minor:	AQL 1.50 percent		
201.	Increments insecurely tied at junctures	Visual/Manual	05005
202.	Straps loose, broken or insecurely tied	Visual/Manual	05006
203.	Marking misleading or unidentifiable	Visual	05007
204.	Evidence of poor workmanship (see 3.6)	Visual	05008

4.2.2.6 Unsealed fiber container (see dwg. 8800641).

Categories	Defects	Method of Inspection	Code No.
Critical: None defined			
Major:	AQL 0.40 percent		
101.	Packing component missing	Visual	06001
102.	Charge inverted	Visual/Manual	06002
Minor:	AQL 1.50 percent		
201.	Charge cannot be removed by hand	Manual	06003
202.	Cover cannot be removed by hand	Manual	06004
203.	Glue or asphalt on round	Visual	06005

4.2.2.7 Sealed wood packing box (see dwg. 8800640).

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Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Contents of box exposed	Visual	07001
102.	DOD symbol misleading or unidentifiable	Visual	07002
103.	Hardware or strapping missing, broken or loose	Visual-Manual	07003
104.	Board broken	Visual	07004
Minor:	AQL 1.50 percent		
201.	Hardware or strapping improperly assembled	Visual-Manual	07005
202.	Car seal missing, unsealed or improperly positioned	Visual-Manual	07006
203.	Handle missing or insecure	Visual-Manual	07007
204.	Marking misleading or unidentifiable	Visual	07008

4.2.3 Testing

4.2.3.1 Moisture

4.2.3.1.1 Black powder.-(see 3.3.1) Major defect - The contractor shall provide adequate controls to insure compliance with the requirements, and shall test for verification, at least one sample of black powder from each eight hours production of igniters. If the sample fails to comply with the applicable requirement the eight hours production represented by the sample shall be rejected. The test shall be performed as specified in 4.3.1.1. Code No 08001.

4.2.3.1.2 Benite strands (see 3.3.2) Major defect.-The contractor shall provide adequate controls to insure compliance with the requirements, and shall test for verification, at least one sample of benite from each eight hours production of igniter core assemblies. If the sample fails to comply with the applicable requirement the eight hours production represented by the sample shall be rejected. The test shall be performed as specified in 4.3.1.2. Code No 08002.

4.2.3.2 Propelling charge increment weight (see 3.4), weight of benite for benite-cellulose nitrate igniter (see dwg. 8863420) and weight of black powder for igniter charges (see dwg. 119176) - Major defects.-The charges shall be weighed and check-weighed 100 percent. The check-weighing shall be accomplished independently of the original weighing using a different balance from that used to make the original weighing and, if performed manually, shall be performed by another operator. Any charge which fails to comply with the applicable requirement shall be classed defective and removed from the lot. Equipment shall be

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in accordance with 4.2.4.

a.	Base igniter	09001
b.	Core igniter	10001
c.	Propellant charge	
	Increment four	11001
	Increment five	11002
	Increment six	11003
	Increment seven	11004

4.2.4 Inspection equipment.-Index of Inspection Equipment Lists Number IEL 119249 identifies the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in paragraph 6.3.

4.2.4.1 Government rights to documentation.-Inspection equipment drawings and lists provided and revised in accordance with the requirements of the IEL may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey and whenever inspection equipment drawings are needed.

4.2.4.2 Supply and maintenance of the equipment listed on the IEL shall be in accordance with Specification MIL-I-45607.

4.3 Test methods and procedures

4.3.1 Moisture

4.3.1.1 Black powder.-The moisture content of the black powder shall be determined in accordance with MIL-P-223. Code No 13001.

4.3.1.2 Benite strands.-The moisture content of the benite strands shall be determined in accordance with MIL-STD-286 method 102.1. Code No 14001.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 Level A.-The propelling charges shall be preserved and packaged in accordance with dwg. 8800641.

5.2 Packing

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5.2.1 Level A.-The propelling charges in the containers shall be packed in accordance with dwg. 8800640.

5.3 Data cards.-Data cards shall be prepared for each lot in accordance with the information specified in Specification MIL-A-2550.

6. NOTES

6.1 Ordering data.-Procurement documents should specify the title, number, and date of this specification.

6.2 Inspection code numbers.-The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

6.3 Inspection equipment.-The contractor shall design inspection equipment as required by the referenced Inspection Equipment Lists in accordance with the instructions of paragraphs 6.3.1 through 6.3.7.

6.3.1 Inspection equipment lists.-Inspection equipment lists (IEL) indicate the availabilty of inspection equipment design by showing in the "number" column of the list of inspection equipment (00 Form 1242-3) the numbers of drawings of existing equipment designs or codes indicating "commercial item" or "contractor design". Design action required of the contractor with respect to the different types of drawings that may be listed is described in paragraphs 6.3.2 and 6.3.3. Action required by the contractor with respect to "commercial item" inspection equipment is described in paragraph 6.3.4. The contractor will be required to prepare drawings for all the equipment coded as "contractor design" in the number column. The code will further indicate whether detailed or specification control drawings are required. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.7.

6.3.2 Ordnance Designs.-Ordnance designs are detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Ordnance design is listed for an item of inspection equipment. Ordnance designs fall into two basic classifications; mandatory and non-mandatory. When an IEL references mandatory Ordnance designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with non-mandatory designs and with the approval of the Government, design alternate inspection equipment to facilitate his operations. Such contractor prepared designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.7.

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6.3.3 Specification control drawings.-Specification control drawings depict the device in outline, descriptive, diagrammatic, or pictorial form only and specify the required performance or other characteristics. Contractors must prepare drawings of their designs in support of specification control drawings. The IEL code will indicate whether drawings must be detailed. These contractor prepared designs must be approved by the Government prior to the fabrication or procuring of the equipment. Commercial equipment meeting the requirements of specification control drawings may be approved if described in sufficient detail to permit identification and evaluation by the Government. Designs shall be submitted for approval as specified in 6.3.7.

6.3.4 Commercial items.-Commercial item inspection equipment is equipment that has a universal application for a specific function. It is comprised of items commonly used by industry and government. Contractors are not required to furnish drawings of commercial item inspection equipment but a list of such equipment must be approved by the Government. Lists shall be submitted for approval to the inspection element of the agency administering the contract.

6.3.5 IEL codes.-The inspection equipment referenced in paragraphs 6.3.1, 6.3.2, 6.3.3, and 6.3.4 will be designated in the IEL by the following codes:

CDOF - Contractor's design responsibility on Ordnance format; ORDM-608-12.
CDCF - Contractor design responsibility on contractor format.
OD - Ordnance design.
ODMU - Ordnance design, mandatory for use.
CE - Commercial equipment.
SCD - Specification control drawing.

6.3.6 Use of standard measuring equipment.-If the contractor desires to use standard measuring equipment in lieu of government or contractor designed gages, diagrammatic sketches or other descriptions of the use of such equipment shall be submitted for approval as specified in 6.3.7.

6.3.7 Submission of contractor designs.-Designs shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND. Design review will normally be accomplished within one month after receipt by Picatinny Arsenal. Partial submission of inspection equipment designs is permissible. However, the Arsenal completion date for design review will be based on the date of the final submission of designs.

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Notice.-When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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Preparing Activity
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Project No. 1320-A042

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